7

Pump Analysis

In this chapter we will discuss centrifugal pumps used in liquid pipelines. Although other types of pumps such as rotary pumps and piston pumps are sometimes used, the majority of pipelines today are operated with single-and multi-stage centrifugal pumps. We will cover the basic design of centrifugal pumps, their performance characteristics and how the performance may be modified by changing pump impeller speeds and trimming impellers. We introduce Affinity Laws for centrifugal pumps, the importance of net positive suction head (NPSH) and how to calculate horsepower requirements when pumping different liquids. Also covered is the pump performance correction for high-viscosity liquids using the Hydraulic Institute chart. We will also illustrate how the performance of two or more pumps in series or parallel configuration can be determined and how to estimate the operating point for a pipeline by using the system head curve.

7.1 Centrifugal Pumps Versus Reciprocating Pumps

Pumps are needed to raise the pressure of a liquid in a pipeline so that the liquid may flow from the beginning of the pipeline to the delivery terminus at the required flow rate and pressure. As the flow rate increases more pump pressure will be required.

Over the years pipelines have been operated with centrifugal as well as reciprocating or positive displacement (PD) pumps. In this chapter we will concentrate mainly on centrifugal pumps, as these are used extensively in most liquid pipelines that transport water, petroleum products, and chemicals. PD pumps are discussed as well, since they are used for liquid injection lines in oil pipeline gathering systems.

Centrifugal pumps develop and convert the high liquid velocity into pressure head in a diffusing flow passage. They generally have a lower efficiency than PD pumps such as reciprocating and gear pumps. However, centrifugal pumps can operate at higher speeds to generate higher flow rates and pressures. Centrifugal pumps also have lower maintenance requirements than PD pumps.

Positive displacement pumps such as reciprocating pumps operate by forcing a fixed volume of liquid from the inlet to the outlet of the pump. These pumps operate at lower speeds than centrifugal pumps. Reciprocating pumps cause intermittent flow. Rotary screw pumps and gear pumps are also PD pumps, but operate continuously unlike reciprocating pumps. Positive displacement pumps are generally larger in size and more efficient compared with centrifugal pumps, but require higher maintenance.

Modern pipelines are mostly designed with centrifugal pumps in preference to PD pumps. This is because there is more flexibility in volumes and pressures with centrifugal pumps. In petroleum pipeline installations where liquid from a field gathering system is injected into a main pipeline, PD pumps may be used.

The performance of a centrifugal pump is depicted as a curve that shows variation in pressure head at different flow rates. The pump head characteristic curve shows the pump head developed on the vertical axis, while the flow rate is shown on the horizontal axis. This curve may be referred to as the H-Q curve or the head-capacity curve. Pump companies use the term capacity when referring to flow rate. Throughout this section we use capacity and flow rate interchangeably when dealing with pumps.

Generally, pump performance curves are plotted for water as the liquid pumped. The head is measured in feet of water and flow rate is shown in gal/min. In addition to the head versus flow rate curve, two other curves are commonly shown on a typical pump performance chart. These are pump efficiency versus flow rate and pump brake horsepower (BHP) versus flow rate. Figure 7.1 shows typical centrifugal pump performance curves consisting of head, efficiency and BHP plotted against flow rate (or capacity). You will also notice another curve referred to as NPSH plotted against flow rate. NPSH will be discussed later in this chapter.

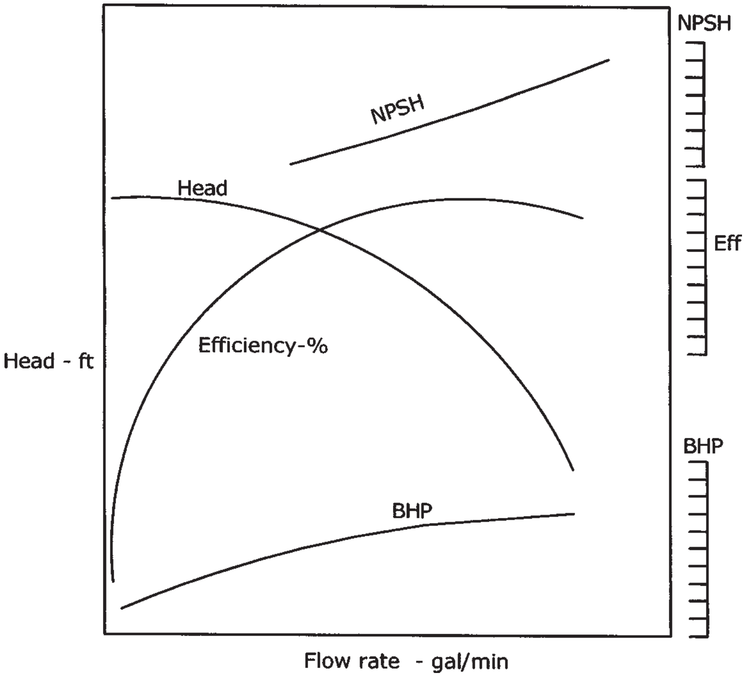

Figure 7.1 Centrifugal pump performance.

In summary, typical centrifugal pump characteristic curves include the following four curves:

Head versus flow rate
Efficiency versus flow rate
BHP versus flow rate
NPSH versus flow rate

The above performance curves for a particular model pump are generally plotted for a particular pump impeller size and speed (example: 10 in. impeller, 3560 RPM).

In addition to the above four characteristic curves, pump head curves drawn for different impeller diameters and iso-efficiency curves are sometimes encountered. When pumps are driven by variable-speed electric motors or engines, the head curves may also be shown at various pump speeds. Pump performance at various impeller sizes and speeds will be discussed in detail later in this chapter.

A PD pump continuously pumps a fixed volume at various pressures. It is able to provide any pressure required at a fixed flow rate. This flow rate

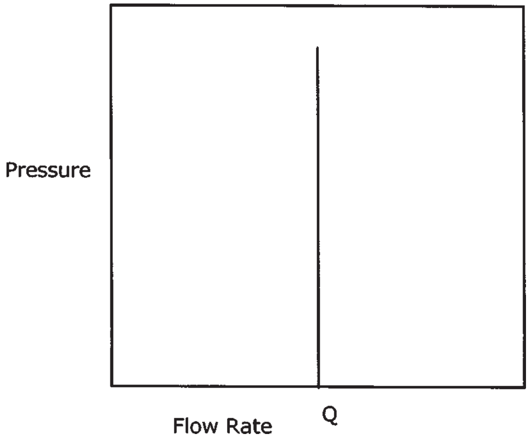

Figure 7.2 Positive displacement pump performance.

depends on the geometry of the pump such as bore and stroke. A typical PD pump pressure-volume curve is shown in Figure 7.2.

7.2 Pump Head Versus Flow Rate

For a centrifugal pump, the head-capacity (flow rate) variation is shown in Figure 7.3.

The head versus flow rate curve (H-Q curve) is generally referred to as a drooping head curve that starts at the highest value (shut-off head) at zero flow rate. The head decreases as the flow rate through the pump increases. The trailing point of the curve represents the maximum flow and the corresponding head that the pump can generate.

Pump head is always plotted in feet of head of water. Therefore the pump is said to develop the same head in feet of liquid, regardless of the liquid. If a particular pump develops 2000 ft head at 3000 gal/min flow rate, we can calculate the pressure developed in psi when pumping water versus another liquid such as gasoline:

With water, pressure developed$=2000\times1/2.31=866$ psi

With gasoline, pressure developed$=2000\times0.74/2.31=641$ psi

This head versus flow rate relationship holds good for this particular pump that has a fixed impeller diameter D and operates at a fixed speed N. The diameter of the impeller generally can be increased within a certain range of sizes, depending on the internal cavity that houses the pump impeller. Thus,

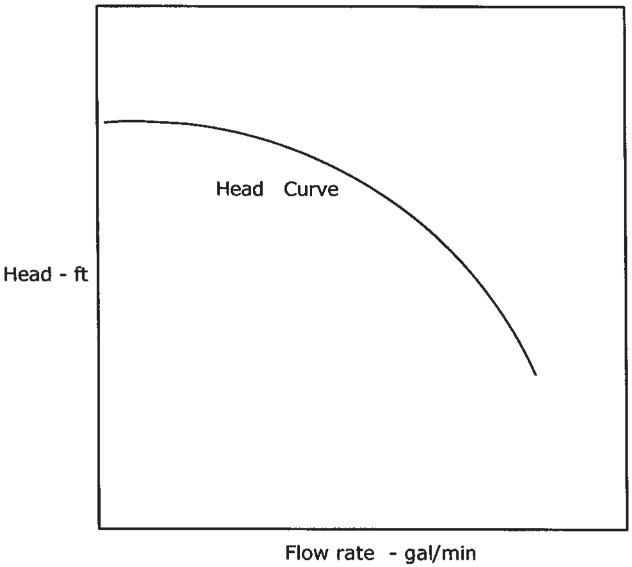

Figure 7.3 Centrifugal pump performance head versus flow rate.

if the H-Q curve in Figure 7.3 is based on a 10 in. impeller diameter, similar curves can be generated for 9 in. impeller and 12 in. impellers representing the minimum and maximum sizes available for this particular pump. The minimum and maximum impeller sizes are defined by the pump vendor for the particular pump model. The 9 in. curve and 12 in. curves would be parallel to the 10 in. curve as shown in Figure 7.4. The variations in pump H-Q curves with impeller diameter follow the Affinity Laws, discussed in a later section of this chapter.

Similar to H-Q variations with pump impeller diameter, curves can be generated for varying pump impeller speeds. If the impeller diameter is kept constant at 10 in. and the initial H-Q curve was based on a pump speed of 3560 RPM (typical induction motor speed), by varying the pump speed we can generate a family of parallel curve as shown in Figure 7.5.

It will be observed that the H-Q variations with pump impeller diameter and pump speed are similar. This is due to the Affinity Laws that the pump performance is based on. We discuss Affinity Laws and prediction of pump performance at different diameters and speeds later in this chapter.

A pump may develop the head in stages. Thus single-stage and multistage pumps are used depending upon the head required. A single-stage

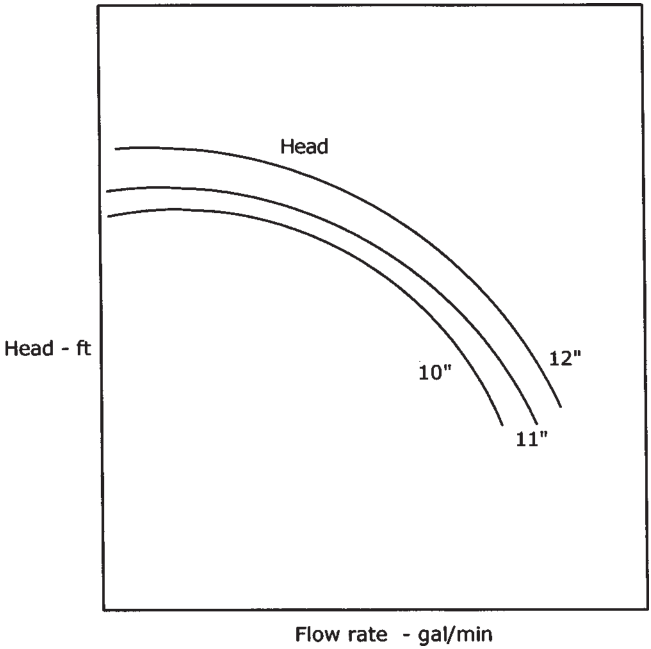

Figure 7.4 Head versus flow rate for different impeller sizes.

pump may generate 200 ft head at 3000 gal/min. A three-stage pump of this design will generate 600 ft head at same flow rate and pump speed. An application that requires 2400 ft head at 3000 gal/min may be served by a multi-stage pump with each stage providing 400 ft of head. De-staging is the process of reducing the active number of stages in a pump to reduce the total head developed. Thus, the six-stage pump discussed above may be de-staged to four stages if we need only 1600 ft head at 3000 gal/min.

7.3 Pump Efficiency Versus Flow Rate

The variation of the efficiency E of a centrifugal pump with flow rate Q is as shown in Figure 7.6. It can be seen that the efficiency starts off at zero value under shut-off conditions (zero flow rate) and rises to a maximum value at some flow rate. After this point, with increase in flow rate the efficiency drops off to some value lower than the peak efficiency. Generally centrifugal pump peak efficiencies are approximately 80–85%. It must be remembered in all these discussions that we are referring to a centrifugal pump

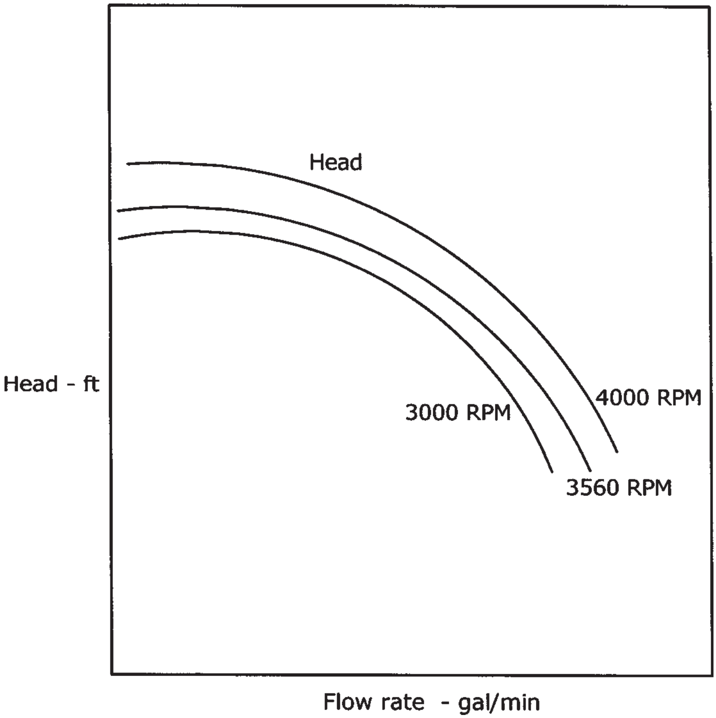

Figure 7.5 Head versus flow rate at different speeds.

performance with water as the liquid being pumped. When heavy, viscous liquids are pumped, these water-based efficiencies will be reduced by a correction factor. Viscosity-corrected pump performance using the Hydraulic Institute charts is discussed in Section 7.8. The flow rate at which the maximum efficiency occurs in a pump is referred to as the best efficiency point (BEP). This flow rate and the corresponding head from the H-Q curve is thus the best operating point on the pump curve since the highest pumping efficiency is realized at this flow rate. In the combined H-Q and efficiency versus flow rate (E-Q) curves shown in Figure 7.7, the BEP is shown as a small triangle.

When choosing a centrifugal pump for a particular application we try to get the operating point as close as possible to the BEP. In order to allow for future increase in flow rates, the initial operating point is chosen slightly to the left of the BEP. This will ensure that with increase in pipeline throughput the operating point on the pump curve will move to the right, which would result in a slightly better efficiency.

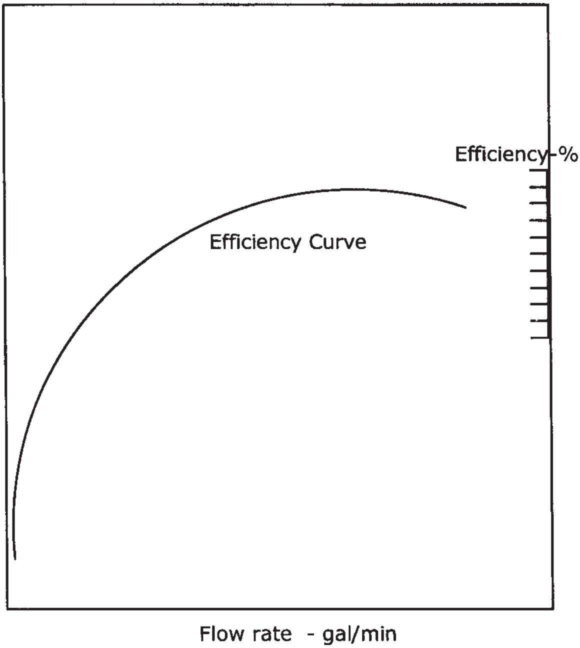

Figure 7.6 Centrifugal pump performance: efficiency versus flow rate.

7.4 BHP Versus Flow Rate

From the H-Q and E-Q curves we can calculate the BHP required at every point along the curve using Equation (5.17) as follows:

$$\text{Pump BHP} = \frac{Q\ H\ Sg}{3960(E)} \tag{7.1}$$

where
Q=Pump flow rate, gal/min
H=Pump head, ft
E=Pump efficiency, as a decimal value less than 1.0
Sg=Liquid specific gravity (for water Sg=1.0)
In SI units, power in kW can be calculated as follows:

$$\text{Power kW} = \frac{Q\ H\ Sg}{367.46\ (E)} \tag{7.2}$$

where
Q=Pump flow rate, m^3/hr
H=Pump head, m

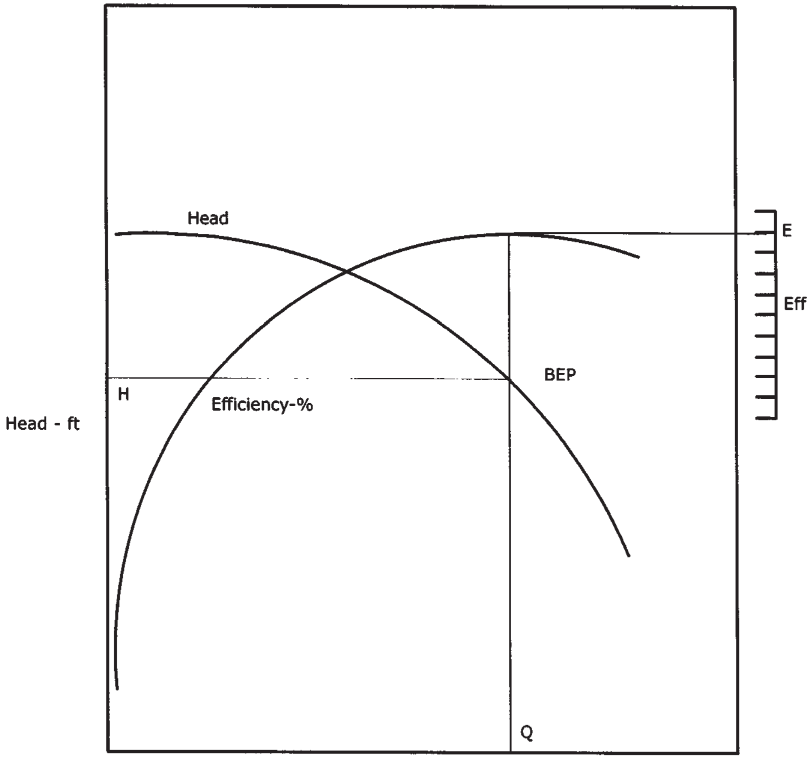

Figure 7.7 Best efficiency point.

E=Pump efficiency, as a decimal value less than 1.0
Sg=Liquid specific gravity (for water Sg=1.0)

For example, if the BEP on a particular pump curve occurs at a flow rate of 3000 gal/min, 2500 ft head at 85% efficiency, the BHP at this flow rate for water is calculated from Equation (7.1) as

$$\text{Pump BHP}=3000\times2500\times1.0/(3960\times0.85)=2228$$

This represents the BHP with water at 3000 gal/min flow rate. Similarly, BHP can be calculated at various flow rates from zero to 4000 gal/min (assumed maximum pump flow) by reading the corresponding head and efficiency values from the H-Q curve and E-Q curve and using Equation (7.1) as above. The BHP versus flow rate can then be plotted as shown in Figure 7.8. The BHP versus flow rate curve can also be shown on the same plot as H-Q and E-Q curves as shown in Figure 7.1.

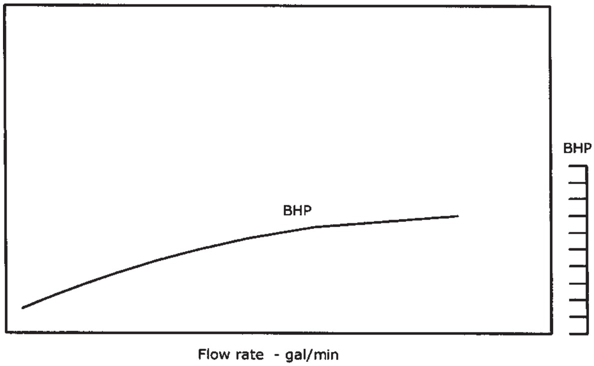

Figure 7.8 BHP versus flow rate.

As discussed in Section 5.6.2, the BHP calculated above is the brake horsepower demanded by the pump. An electric motor driving the pump will have an efficiency that ranges from 95% to 98%. Therefore the electric motor HP required is the pump BHP divided by the motor efficiency as follows

Motor HP=Pump BHP/Motor efficiency
=2228/0.96
=2321 HP, based on 96% motor efficiency

In determining the motor size required, we must first calculate the maximum HP demanded by the pump, which will generally (but not necessarily) be at the highest flow rate. The BHP at this flow rate will then be divided by the electric motor efficiency to get the maximum motor HP required. The next available standard size motor will be selected for the application.

Suppose that for the above pump 4000 gal/min is the maximum flow rate through the pump and the head and efficiency are 1800 ft and 76%, respectively. We will calculate the maximum BHP required as

$$\text{Maximum pump BHP} = \frac{4000 \times 1800 \times 1.0}{3960 \times 0.76} = 2392.34$$

$$\text{Motor HP} = \frac{2392.34}{0.96} = 2492$$

Therefore a 2500 HP electric motor will be adequate for this case. Before we leave the subject of pump BHP and electric motor HP, we must mention that electric motors have built into their nameplate rating a service factor. The service factor for most induction motors is in the range of 1.10 to 1.15. This simply means that if the motor's rated HP is 2000, the 1.15 service factor allows the motor to supply up to a maximum of 1.15×2000 or 2300 HP without burning up the motor windings. However, it is not advisable to use this extra 300 HP on a continuous basis, as explained below.

At a 1.15 service factor rating, the motor windings will be taking on an additional 15% electric current compared with their rated current (amps). This translates to extra heat that will shorten the life of the windings. Even though the extra current is only 15%, the heat generated will be over 32%, since electrical heating is proportional to the square of the current ($1.15^2=1.3225$). However, while continuous operation at the service factor rating is discouraged, in an emergency this additional 15% motor HP above the rated HP may be used.

7.5 NPSH Versus Flow Rate

In addition to the three pump curves discussed previously—for head versus capacity (H-Q), efficiency versus capacity (E-Q), and BHP versus capacity (BHP-Q)—the pump performance data will include a fourth curve for net positive suction head (NPSH) versus capacity. This curve is generally located above the head, efficiency, and BHP curves as shown in Figure 7.1.

The NPSH curve shows the variation in the minimum net positive suction head at the impeller suction versus the flow rate. The NPSH increases at a faster rate as the flow rate increases. NPSH is defined as the net positive suction head required at the pump impeller suction to prevent pump cavitation at any flow rate. It represents the resultant positive pressure at pump suction after accounting for frictional loss and liquid vapor pressure. NPSH is discussed in detail in Section 7.12. At present, we can say that as pump flow rate increases the NPSH required also increases. We will perform calculations to compare NPSH required (per pump curve) versus actual NPSH available based on pump suction piping and other parameters that affect the available suction pressure at the pump.

7.6 Specific Speed

The specific speed of a centrifugal pump is a parameter based on the impeller speed, flow rate, and head at the best efficiency. It is used for comparing geometrically similar pumps and for classifying the different types of centrifugal pumps.

Specific speed may be defined as the speed at which a geometrically similar pump must be run such that it produces a head of 1 ft at a flow rate of 1 gal/min. Mathematically, specific speed is defined as

$$N_S = NQ^{1/2}/H^{3/4} \tag{7.3}$$

where

N_S=Pump specific speed
N=Pump impeller speed, RPM
Q=Flow rate or capacity, gal/min
H=Head, ft

Both Q and H are measured at the BEP for the maximum impeller diameter. The head H is measured per stage for a multi-stage pump. Low specific speed is associated with high-head pumps, while high specific speed is found with low-head pumps.

Another related term, the suction specific speed, is defined as follows:

$$N_{SS} = NQ^{1/2}/(NPSH_R)^{3/4} \tag{7.4}$$

where

N_{SS}=Suction specific speed
N=Pump impeller speed, RPM
Q=Flow rate or capacity, gal/min
$NPSH_R$=NPSH required at the BEP

When applying the above equations to calculate the pump specific speed and suction specific speed, use the full Q value for single or double suction pumps for N_S calculation. For N_{SS} calculation, use one-half the Q value for double suction pumps.

Example Problem 7.1

Calculate the specific speed of a five-stage double suction centrifugal pump, 12 in. diameter impeller, that when operated at 3560 RPM generates a head of 2200 ft at a capacity of 3000 gal/min at the BEP on the head capacity curve. If the NPSH required is 25 ft, calculate the suction specific speed.

Solution

$$N_S = NQ^{1/2}/H^{3/4}$$

$$=3560(3000)^{1/2}/(2200/5)^{3/4}=2030$$

The suction specific speed is

$$N_{SS} = N\,Q^{1/2}/NPSH_R^{3/4}$$

$$= 3560(3000/2)^{1/2}/(25)^{3/4} = 12{,}332$$

Table 7.1 Specific Speeds of Centrifugal Pumps

Description	Application	Specific speed, Ns
Radial vane	Low capacity, high head	500–1000
Francis-screw type	Medium capacity, medium head	1000–4000
Mixed-flow type	Medium to high capacity, low to medium head	4000–7000
Axial-flow type	High capacity, low head	7000–20,000

Centrifugal pumps are generally classified as

(a) Radial flow
(b) Axial flow
(c) Mixed flow

Radial flow pumps develop head by centrifugal force. Axial flow pumps on the other hand develop the head due to the propelling or lifting action of the impeller vanes on the liquid. Radial flow pumps are used when high heads are required, while axial flow and mixed flow pumps are mainly used for low-head/high-capacity applications. Table 7.1 lists the specific speed range for centrifugal pumps.

7.7 Affinity Laws: Variation with Impeller Speed and Diameter

Each family of pump performance curves for head, efficiency, and BHP versus capacity is specific to a particular size of impeller and pump model. Thus, Figure 7.1 depicts pump performance data for a 10 in. diameter pump impeller. This particular pump may be able to accommodate a larger impeller, such as 12 in. in diameter. Alternatively, a smaller, 9 in. impeller may also be fitted in this pump casing. The range of impeller sizes is dependent on the pump case design and will be defined by the pump vendor. Since centrifugal pumps generate pressure due to centrifugal force, it is clear that a smaller impeller diameter will generate less pressure at a given flow rate than a larger impeller at the same speed. If the performance data for a 10 in. impeller is available, we can predict the pump performance when using a larger or smaller size impeller by means of the centrifugal pump Affinity Laws.

The same pump with a 10 in. impeller may be operated at a higher or lower speed to provide higher or lower pressure. Most centrifugal pumps are driven by constant-speed electric motors. A typical induction motor in the United States operates at 60 Hz and will have a synchronous speed of

3600 RPM. With some slip, the induction motor would probably run at 3560 RPM. Thus, most constant-speed motor-driven pumps have performance curves based on 3560 RPM. Some slower-speed pumps operate at 1700 to 1800 RPM. If a variable-speed motor is used, the pump speed can be varied from, say, 3000 to 4000 RPM. This variation in speed produces variable head versus capacity values for the same 10 in. impeller. Given the performance data for a 10 in. impeller at 3560 RPM, we can predict the pump performance at different speeds ranging from 3000 to 4000 RPM using the centrifugal pump Affinity Laws as described next.

The Affinity Laws for centrifugal pumps are used to predict pump performance for changes in impeller diameter and impeller speed. According to the Affinity Laws, for small changes in impeller diameter the flow rate (pump capacity) Q is directly proportional to the impeller diameter. The pump head H, on the other hand, is directly proportional to the square of the impeller diameter. Since the BHP is proportional to the product of flow rate and head (see Equation 7.1), BHP will vary as the third power of the impeller diameter.

The Affinity Laws are represented mathematically as follows:

For impeller diameter change:

$$Q_2/Q_1=D_2/D_1 \tag{7.5}$$

$$H_2/H_1=(D_2/D_1)^2 \tag{7.6}$$

where

Q_1, Q_2=Initial and final flow rates
H_1, H_2=Initial and final heads
D_1, D_2=Initial and final impeller diameters

From the H-Q curve corresponding to the impeller diameter D_1 we can select a set of H-Q values that cover the entire range of capabilities from zero to the maximum possible. Each value of Q and H on the corresponding H-Q curve for the impeller diameter D_2 can then be computed using the Affinity Law ratios per Equations (7.5) and (7.6).

Similarly, Affinity Laws state that for the same impeller diameter, if pump speed is changed, flow rate is directly proportional to the speed, while the head is directly proportional to the square of the speed. As with diameter change, the BHP is proportional to the third power of the impeller speed. This is represented mathematically as follows:

For impeller speed change:

$$Q_2/Q_1=N_2/N_1 \tag{7.7}$$

$$H_2/H_1=(N_2/N_1)^2 \tag{7.8}$$

where

Q_1, Q_2=Initial and final flow rates
H_1, H_2=Initial and final heads
N_1, N_2=Initial and final impeller speeds

Note that the Affinity Laws for speed change are exact. However, the Affinity Laws for impeller diameter change are only approximate and valid for small changes in impeller sizes. The pump vendor must be consulted to verify that the predicted values using Affinity Laws for impeller size changes are accurate or whether any correction factors are needed. With speed and impeller size changes, the efficiency versus flow rate can be assumed to be the same.

An example using the Affinity Laws will illustrate how the pump performance can be predicted for impeller diameter change as well as for impeller speed change.

Example Problem 7.2

The head and efficiency versus capacity data for a centrifugal pump with a 10 in. impeller is as shown below.

Q, gal/min	0	800	1600	2400	3000
H, ft	3185	3100	2900	2350	1800
E, %	0.0	55.7	78.0	79.3	72.0

The pump is driven by a constant-speed electric motor at a speed of 3560 RPM.

(a) Determine the performance of this pump with an 11 in. impeller, using Affinity Laws.
(b) If the pump drive were changed to a variable frequency drive (VFD) motor with a speed range of 3000 to 4000 RPM, calculate the new H-Q curve for the maximum speed of 4000 RPM with the original 10 in. impeller.

Solution

(a) Using Affinity Laws for impeller diameter changes, the multiplying factor for flow rate is factor=11/10=1.1 and the multiplier for head is $(1.1)^2$=1.21. Therefore, we will generate a new set of Q and H values for the 11 in. impeller by multiplying the given Q values by the factor 1.1 and the H values by the factor 1.21 as follows:

Q, gal/min	0	880	1760	2640	3300
H, ft	3854	3751	3509	2844	2178

The above flow rate and head values represent the predicted performance of the 11 in. impeller. The efficiency versus flow rate curve for the 11 in. impeller will be approximately the same as that of the 10 in. impeller.

(b) Using Affinity Laws for speeds, the multiplying factor for the flow rate is factor=4000/3560=1.1236 and the multiplier for head is $(1.1236)^2$=1.2625. Therefore we will generate a new set of Q and H values for the pump at 4000 RPM by multiplying the given Q values by the factor 1.1236 and the H values by the factor 1.2625 as follows:

Q, gal/min	0	899	1798	2697	3371
H, ft	4021	3914	3661	2967	2273

The above flow rates and head values represent the predicted performance of the 10 in. impeller at 4000 RPM. The new efficiency versus flow rate curve will be approximately the same as the given curve for 3560 RPM.

Thus, using Affinity Laws, we can determine whether we should increase or decrease the impeller diameter to match the requirements of flow rate and head for a specific pipeline application.

For example, suppose hydraulic calculations for a particular pipeline indicate we need a pump that can provide a head of 2000 ft at a flow rate of 2400 gal/min. The data for the pump in Example Problem 7.2 shows that a 10 in. impeller produces 2350 ft head at a flow rate of 2400 gal/min. If this is a constant-speed pump, it is clear that in order to get 2000 ft head, the current 10 in. impeller needs to be trimmed in diameter to reduce the head from 2350 ft to the 2000 ft required. The Example Problem 7.4 below will illustrate how the amount of impeller trim is calculated.

7.8 Effect of Specific Gravity and Viscosity on Pump Performance

The performance of a pump as reported by the pump vendor is always based on water as the pumped liquid. Hence the head versus capacity curve, the efficiency versus capacity curve, BHP versus capacity curve, and NPSH required versus capacity curve are all applicable only when the liquid pumped is water (specific gravity of 1.0) at standard conditions, usually 60°F. When pumping a liquid such as crude oil with specific gravity of 0.85 or a refined product such as gasoline with a specific gravity of 0.74, both the H-Q and E-Q curves may still apply for these products if the viscosities are sufficiently low (less than 4.3 cSt for small pumps up to 100 gal/min capacity and less than 40 SSU for larger pumps up to 10,000 gal/min) according to Hydraulic Institute standards. Since BHP is a function of the specific gravity, from Equation (7.1) it is clear that the BHP versus flow rate will be different for other liquids compared with water. If the liquid pumped has a higher viscosity, in the range of 4.3 to 3300 cSt, the head, efficiency, and BHP curves based on water must all be corrected for the high viscosity. The Hydraulic Institute has published viscosity correction charts that can be applied to correct the water performance curves to produce viscosity-corrected curves. See Figure 7.9 for details of this chart. For any application involving high-viscosity liquids, the pump vendor should be given the liquid properties. The viscosity-corrected performance curves will be supplied by the vendor as part of the pump proposal. As an end user you may also use these charts to generate the viscosity-corrected pump curves.

The Hydraulic Institute method of viscosity correction requires determining the BEP values for Q, H, and E from the water performance curve. This is called the 100% BEP point. Three additional sets of Q, H, and E values are obtained at 60%, 80%, and 120% of the BEP flow rate from the water performance curve. From these four sets of data, the Hydraulic Institute chart can be used to obtain the correction factors C_q, C_h, and C_e for flow, head, and efficiency for each set of data. These factors are used to multiply the Q, H, and E values from the water curve, thus generating corrected values of Q, H, and E for 60%, 80%, 100%, and 120% BEP values. Example Problem 7.3 illustrates the Hydraulic Institute method of viscosity correction. Note that for multi-stage pumps, the values of H must be per stage.

Example Problem 7.3

The water performance of a single-stage centrifugal pump for 60%, 80%, 100%, and 120% of the BEP is as shown below:

Q, gal/min	450	600	750	900
H, ft	114	108	100	86
E, %	72.5	80.0	82.0	79.5

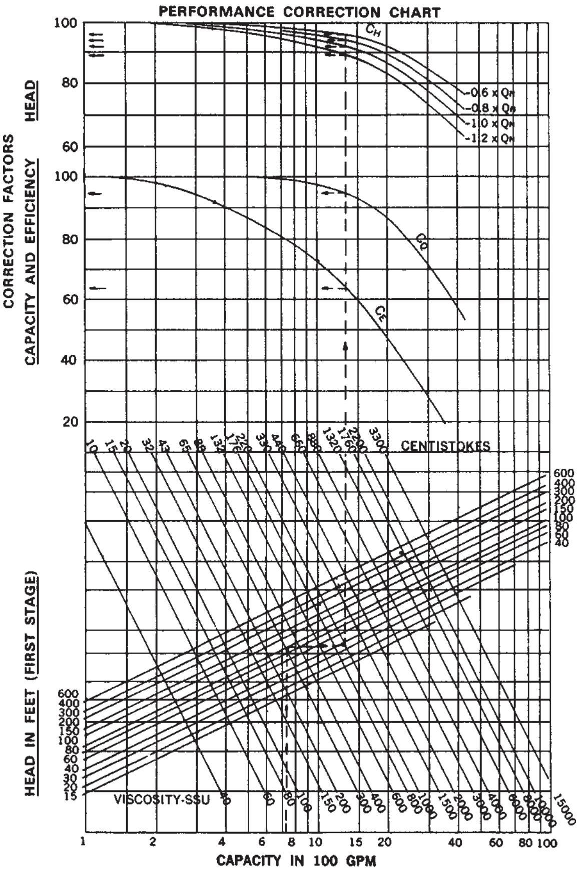

Figure 7.9 Viscosity correction chart. (Courtesy of Hydraulic Institute, Parsippany, NJ; www.pumps.org.)

Calculate the viscosity-corrected pump performance when pumping oil with a specific gravity of 0.90 and a viscosity of 1000 SSU at pumping temperature.

Solution

By inspection, the BEP for this pump curve is

Q=750
H=100
E=82

We first establish the four sets of capacities to correspond to 60%, 80%, 100%, and 120%. These have already been given as 450, 600, 750, and

900 gal/min. Since the head values are per stage, we can directly use the BEP value of head along with the corresponding capacity to enter the Hydraulic Institute viscosity correction chart at 750 gal/min on the lower horizontal scale, go vertically from 750 gal/min to the intersection point on the line representing the 100 ft head curve and then horizontally to intersect the 1000 SSU viscosity line, and finally vertically up to intersect the three correction factor curves C_e, C_q, and C_h.

From the Hydraulic Institute chart of correction factors (Figure 7.9) we obtain the values of C_q, C_h, and C_e for flow rate, head, and efficiency as follows:

C_q	0.95	0.95	0.95	0.95
C_h	0.96	0.94	0.92	0.89
C_e	0.635	0.635	0.635	0.635

corresponding to Q values of 450 (60% of Q_{NW}), 600 (80% of Q_{NW}), 750 (100% of Q_{NW}), and 900 (120% of Q_{NW}). The term Q_{NW} is the BEP flow rate from the water performance curve.

Using these correction factors, we generate the Q, H, and E values for the viscosity-corrected curves by multiplying the water performance value of Q by C_q, H by C_h, and E by C_e and obtain the following result:

Q_V	427	570	712	855
H_V	109.5	101.5	92.0	76.5
E_V	46.0	50.8	52.1	50.5
BHP_V	23.1	25.9	28.6	29.4

The last row of values for viscous BHP was calculated using Equation (7.1) as follows:

$$BHP_V=(Q_V)(H_V)0.9/(39.60\times E_V) \tag{7.9}$$

where Q_V, H_V, and E_V are the viscosity-corrected values of capacity, head, and efficiency tabulated above.

Note that when using the Hydraulic Institute chart for obtaining the correction factors, two separate charts are available. One chart applies to small pumps up to 100 gal/min capacity and head per stage of 6 to 400 ft. The other chart applies to larger pumps with capacity between 100 and 10,000 gal/min and head range of 15 to 600 ft per stage. Also, remember that when data is taken from a water performance curve, the head has to be corrected per stage, since the Hydraulic Institute charts are based on head in ft per stage rather than the total pump head. Therefore, if a six-stage pump has a BEP at 2500 gal/min and 3000 ft of head with an efficiency of 85%, the head per stage to be used with the chart will be 3000 divided by 6=500 ft. The total head (not per stage) from the water curve can then be multiplied by the correction factors from the Hydraulic Institute charts to obtain the viscosity-corrected head for the six-stage pump.

7.9 Pump Curve Analysis

So far we have discussed the performance of a single pump. When transporting liquid through a pipeline we may need to use more than one pump at a pump station to provide the necessary flow rate or head requirement. Pumps may be operated in series or parallel configurations. Series pumps are generally used for higher heads and parallel pumps for increased flow rates.

For example, let us assume that pressure drop calculations indicate that the originating pump station on a pipeline requires 900 psi differential pressure to pump 3000 gal/min of gasoline with specific gravity 0.736. Converting to customary pump units we can state that we require a pump that can provide the following:

$$\text{Head} = \frac{900 \times 2.31}{0.736} = 2825 \text{ ft}$$

at 3000 gal/min flow rate. We have two options here. We could select from a pump vendors catalog one large pump that can provide 2825 ft head at 3000 gal/min or select two smaller pumps that can provide 1413 ft at 3000 gal/ min. We would use these two smaller pumps in series to provide the required head, since for pumps operated in series the same flow rate goes through each pump and the resultant head is the sum of the heads generated by each pump. Alternatively, we could select two pumps that can provide 2825 ft at 1500 gal/min each. In this case these pumps would be operated in parallel. In parallel operation, the flow rate is split between the pumps, while each pump produces the same head. Series and parallel pump configuration are illustrated in Figure 7.10.

The choice of series or parallel pumps for a particular application depends on many factors, including pipeline elevation profile, as well as operational flexibility. Figure 7.11 shows the combined performance of two identical pumps in series, versus parallel configuration. It can be seen that parallel pumps are used when we need larger flows. Series pumps are used when we need higher heads than each individual pump. If the pipeline elevation profile is essentially flat, the pump pressure is required mainly to overcome the pipeline friction. On the other hand, if the pipeline has drastic elevation changes, the pump head generated is mainly for the static

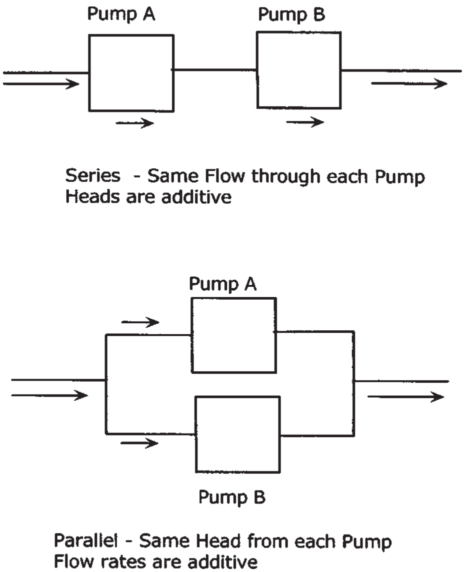

Figure 7.10 Pumps in series and parallel.

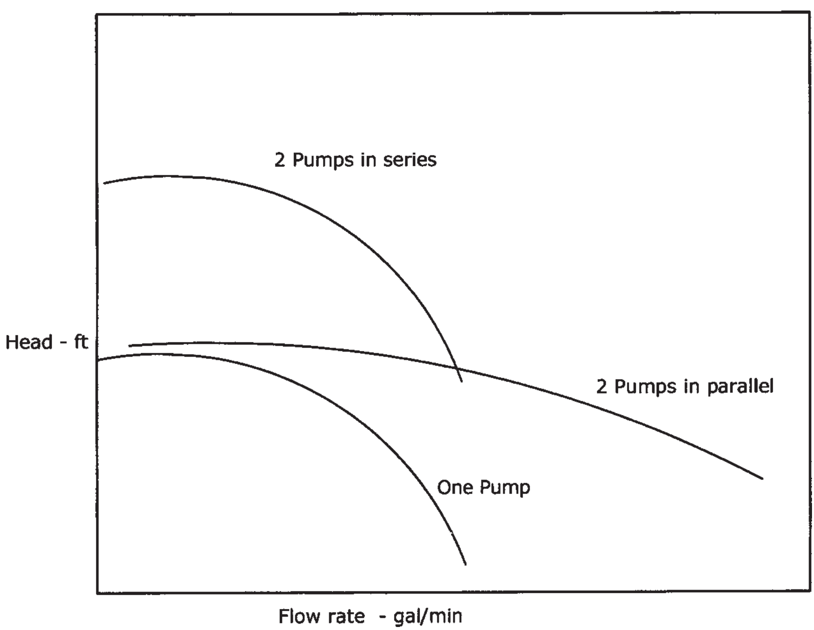

Figure 7.11 Pump performance: series and parallel.

lift and to a lesser extent for pipe friction. In the latter case, if two pumps are used in series and one shuts down, the remaining pump alone will only be able to provide half the head and therefore will not be able to provide the necessary head for the static lift at any flow rate. If the pumps were configured in parallel, then shutting down one pump will still allow the other pump to provide the necessary head for the static lift at half the previous flow rate. Thus parallel pumps are generally used when elevation differences are considerable. Series pumps are used where pipeline elevations are not significantly high.

Example Problem 7.4

One large pump and one small pump are operated in series. The H-Q characteristics of the pumps are defined as follows:

Pump 1

Q, gal/min	0	800	1600	2400	3000
H, ft	2389	2325	2175	1763	1350

Pump 2

Q, gal/min	0	800	1600	2400	3000
H, ft	796	775	725	588	450

(a) Calculate the combined performance of pump 1 and pump 2 in series configuration.
(b) What changes (trimming impellers) must be made to either pump to satisfy the requirement of 2000 ft of head at 2400 gal/min when operated in series?
(c) Can these pumps be configured to operate in parallel?

Solution

(a) Pumps in series have the same flow through each pump and the heads are additive. We can therefore generate the total head produced in series configuration by adding the head of each pump for each flow rate given as follows:

Combined performance of pump 1 and pump 2 in series:

Q, gal/min	0	800	1600	2400	3000
H, ft	3185	3100	2900	2351	1800

(b) It can be seen from the above combined performance that the head generated at 2400 gal/min is 2351 ft. Since the design requirement is 2000 ft at this flow rate, it is clear that the head needs to be reduced by trimming one of the pump impellers to produce the necessary total head. We will proceed as follows.

Let us assume that the smaller pump will not be modified and the impeller of the larger pump (pump 1) will be trimmed to produce the necessary head.

Modified head required of pump 1=2000-588=1412 ft.

Pump 1 produces 1763 ft of head at 2400 gal/min. In order to reduce this head to 1412 ft, we must trim the impeller by approximately

$(1412/1763)^{1/2}$=0.8949 or 89.5%

based on Affinity Laws. This is only approximate and we need to generate a new H-Q curve for the trimmed pump 1 so we can verify that the desired head will be generated at 2000 gal/min. Using the method in Example Problem 7.2 we can generate a new H-Q curve for a trim of 89.5%:

Flow multiplier=0.8949
Head multiplier=$(0.8949)^2$=0.8008

Pump 1 trimmed to 89.5% of present impeller diameter:

Q, gal/min	0	716	1432	2148	2685
H, ft	1913	1862	1742	1412	1081

It can be seen from above trimmed pump performance that the desired head of 1412 ft will be achieved at a flow rate of 2148 gal/min. Therefore, at the lower flow rate of 2000 gal/min, we can estimate by interpolation that the head would be higher than 1412 ft. Hence, slightly more trimming would be required to achieve the design point of 1412 ft at 2000 gal/min. By trial and error we arrive at a pump trim of 87.7% and the resulting pump performance for pump 1 at 87.7% trim is as follows:

Pump 1 trimmed to 87.7% of present impeller diameter:

Q, gal/min	0	702	1403	2105	2631
H, ft	1837	1788	1673	1356	1038

By plotting this curve, we can verify that the required head of 1412 ft will be achieved at 2000 gal/min.

(c) To operate satisfactorily in a parallel configuration, the two pumps must have a common range of heads so that at each common head, the corresponding flow rates can be added to determine the combined performance. Pump 1 and pump 2 are mismatched for parallel operation. Therefore, they cannot be operated in parallel.

7.10 Pump Head Curve Versus System Head Curve

In Chapter 5 we discussed the development of pipeline system head curves. In this section we will see how the system head curve together with the pump head curve can predict the operating point (flow rate Q—head H) on the pump curve.

Since the system head curve for the pipeline is a graphic representation of the pressure required to pump a product through the pipeline at various flow rates (increasing pressure with increasing flow rate) and the pump H-Q curve shows the pump head available at various flow rates, when the head requirements of the pipeline match the available pump head we have a point of intersection of the system head curve with the pump head curve as shown in Figure 7.12. This is the operating point for this pipe-pump combination.

The point of intersection of the pump head curve and the system head curve for diesel (point A) indicates the operating point for this pipeline

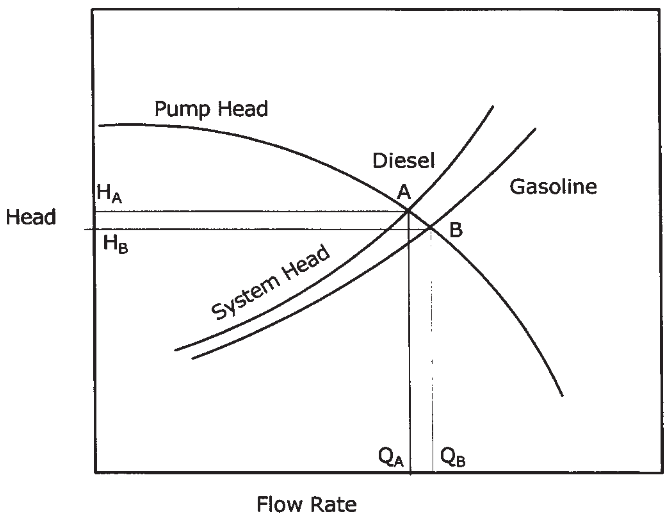

Figure 7.12 Pump head and system head curves.

with diesel. Similarly, if gasoline were pumped through this pipeline, the corresponding operating point (point B) is as shown in Figure 7.12. Therefore, with 100% diesel in the pipeline, the flow rate would be Q_A and the corresponding pump head would be H_A as shown. Similarly, with 100% gasoline in the pipeline, the flow rate would be Q_B and the corresponding pump head would be H_B as shown in Figure 7.12. When batching the two products, a certain proportion of the pipeline will be filled with diesel and the rest will be filled with gasoline. The flow rate will then be at some point between Q_A and Q_B, since a new system head curve located between the diesel curve and the gasoline curve will dictate the operating point.

If we had plotted the system head curve in psi instead of ft of liquid head, the Y-axis will be the pressure required (psi). The pump head curve also needs to be converted to psi versus flow rate by using Equation (3.7):

$$\text{Psi} = \text{Head} \times \text{Sg}/2.31$$

Therefore, using the same pump we will have two separate H-Q curves for diesel and gasoline, due to different specific gravities. Such a situation is shown in Figure 7.13. Even though the pump develops the same head (in ft) with diesel or gasoline (or water), the pressure generated in psi will be different, and hence the two separate H-Q curves, as indicated in Figure 7.13.

In a batched pipeline, the operating point moves from D to A, A to B, B to C, and C to D as shown in Figure 7.13. We start off with 100% diesel in

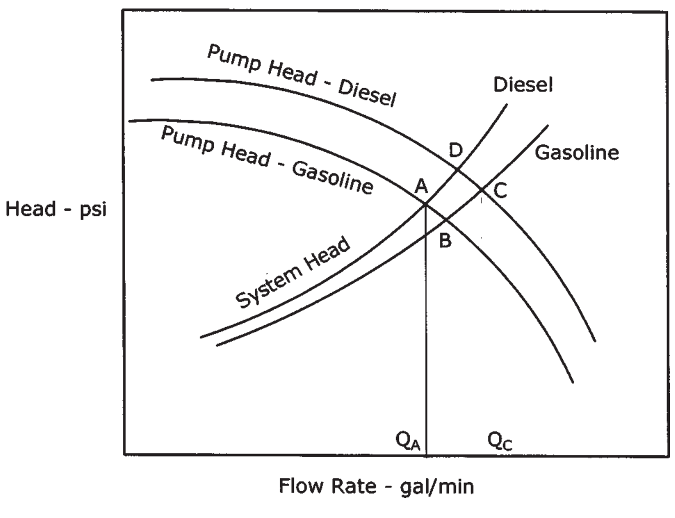

Figure 7.13 Pump and system curve: batching.

the pipeline and the pump. This is point D in the figure. When gasoline enters the pump, with diesel in the pipeline, the operating point moves from D to A. Then, as the gasoline batch enters the pipeline, the system head curve moves to the right until it reaches point B representing the operating point with 100% gasoline in the pipeline and gasoline in the pump. As diesel reaches the pump and the line is still full of gasoline, the operating point moves to C, where we have diesel in the pump and the pipeline filled with gasoline. Finally, as the diesel batch enters the pipeline, the operating point moves towards D. At D we have completed the cycle and both the pump and the pipeline are filled with diesel. Thus it is seen that in a batched operation the flow rates vary between Q_A and Q_C as in Figure 7.13.

7.11 Multiple Pumps Versus System Head Curve

When two or more pumps are operated in parallel on a pipeline system, we saw how the pump head curves added the flow rates at the same head to create the combined pump performance curve. Similarly, with series pumps, the heads are added up for the same flow rate resulting in the combined pump head curve.

Figure 7.14 illustrates the pipeline system head curve superimposed on the pump head curves to show the operating point with one pump, two

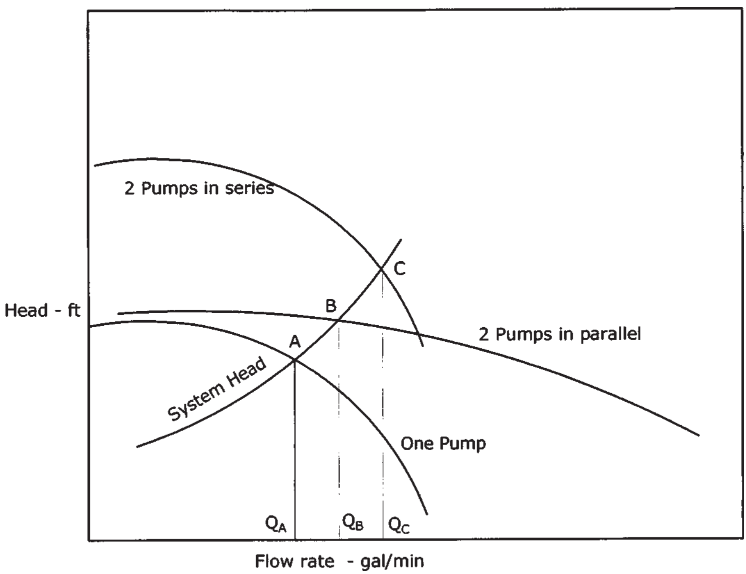

Figure 7.14 Multiple pumps and system head curve.

pumps in series and the same two pumps in parallel configurations. The operating points are shown as A, C, and B with flow rates of Q_A, Q_C, and Q_B, respectively.

In certain pipeline systems, depending upon the flow requirements, we may be able to obtain higher throughput by switching from a series pump configuration to a parallel pump configuration. From Figure 7.14 it can be seen that a steep system head curve would favor pumps in series, while a relatively flat system head curve is associated with the operation of parallel pumps.

7.12 NPSH Required Versus NPSH Available

As the pressure on the suction side of a pump is reduced to a value below the vapor pressure of the liquid being pumped, flashing can occur. The liquid vaporizes and the pump is starved of liquid. At this point the pump is said to cavitate due to insufficient liquid volume and pressure. The vapor can damage the pump impeller, further reducing its ability to pump. To avoid vaporization of liquid, we must provide adequate positive pressure at the pump suction that is greater than the liquid vapor pressure.

NPSH for a centrifugal pump is defined as the net positive suction head required at the pump impeller suction to prevent pump cavitation at any flow rate. Cavitation will damage the pump impeller and render it useless. NPSH represents the resultant positive pressure at the pump suction. In this section, we will analyze a piping configuration from a storage tank to a pump suction, to calculate the available NPSH and compare it with the NPSH required by the pump vendor's performance curve. The NPSH available will be calculated by taking into account any positive tank head, including atmospheric pressure, and subtracting the pressure drop due to friction in the suction piping and the liquid vapor pressure at the pumping temperature. The resulting value of NPSH for this piping configuration will represent the net pressure of the liquid at pump suction, above its vapor pressure. The value calculated must be more than the NPSH specified by the pump vendor at the particular flow rate.

Before we calculate the NPSH available in a typical pump-piping configuration, let us analyze the piping geometry associated with a pump taking suction from a tank and delivering liquid to another tank as shown in Figure 7.15.

The vertical distance from the liquid level on the suction side of the pump center line is defined as the static suction head. More correctly, it is the static suction lift (H_S) when the center line of the pump is above that of the liquid supply level as depicted in Figure 7.15. If the liquid supply level is higher than the pump center line, it is called the static suction head on

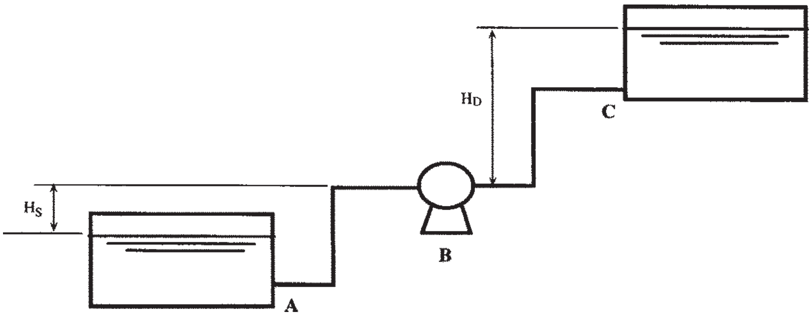

Figure 7.15 Centrifugal pump: suction and discharge heads.

the pump. Similarly, the vertical distance from the pump center line to the liquid level on the delivery side is called the static discharge head (H_D) as shown in Figure 7.15.

The total static head on a pump is defined as the sum of the static suction head and the static discharge head. It represents the vertical distance between the liquid supply level and the liquid discharge level. The static suction head, static discharge head, and the total static head on a pump are all measured in feet of liquid, or meters of liquid in the SI system.

The friction head, measured in feet of liquid pumped, represents the pressure drop due to friction in both suction and discharge piping. It represents the pressure required to overcome the frictional resistance of all piping, fittings, and valves on the suction side and discharge side of the pump as shown in Figure 7.15. Reviewing the piping system shown in Figure 7.15, it is seen that there are three sections of straight piping and two pipe elbows on the suction side between the liquid supply (A) and the center line of the pump. In addition, there would be at least two valves, one at the tank and the other at the inlet to the pump suction. An entrance loss will be added to account for the pipe entrance at the tank.

Similarly, on the discharge side of the pump between the center line of the pump (B) and the tank delivery point (C) there are three straight sections of pipe and two pipe elbows along with two valves. On the discharge of the pump there would also be a check valve to prevent reverse flow through the pump. An exit loss at the tank entry will also be added to account for the delivery pipe.

On the suction side of the pump, the available suction head H_S will be reduced by the friction loss in the suction piping. This net suction head on the pump will be the available suction head at the pump center line.

On the discharge side the discharge head H_D will be increased by the friction loss in the discharge piping. This is the net discharge head on the pump.

Mathematically,

$$\text{Suction head}=H_S-H_{fs} \tag{7.10}$$

$$\text{Discharge head}=H_D+H_{fd} \tag{7.11}$$

where

H_{fs}=Friction loss in suction piping
H_{fd}=Friction loss in discharge piping

If the suction piping is such that there is a suction lift (instead of suction head) the value of H_S in Equation (7.10) will be negative. Thus a 20 ft static suction lift combined with a suction piping loss of 2 ft will actually result in an overall suction lift of 22 ft. The discharge head, on the other hand, will be the sum of the discharge head and friction loss in the discharge piping. Assuming a 30 ft discharge head and 5 ft friction loss, the total discharge head will be 30+5=35 ft. In this example, the total head of the pump is 22+ 35=57 ft.

Example Problem 7.5

A centrifugal pump is used to pump a liquid from a storage tank through 500 ft of suction piping as shown in Figure 7.16.

(a) Calculate the NPSH available at a flow rate of 3000 gal/min.
(b) The pump vendor's data indicate the NPSH required to be 35 ft at 3000 gal/min and 60 ft at 4000 gal/min. Can this piping system handle the higher flow rate without the pump cavitating?
(c) If cavitation is a problem in (b) above, what changes must be made to the piping system to prevent pump cavitation at 4000 gal/min?

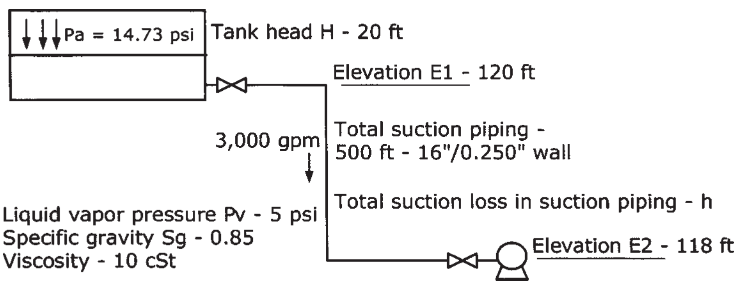

Figure 7.16 NPSH calculation.

Solution

(a) NPSH available in ft of liquid head:

$$(P_a\text{-}P_v)(2.31/Sg)+H+E1\text{-}E2\text{-}h \tag{7.12}$$

where

P_a=Atmospheric pressure, psi
P_v=Liquid vapor pressure at the flowing temperature, psi
Sg=Liquid specific gravity
H=Tank head, ft
E1=Elevation of tank bottom, ft
E2=Elevation of pump suction, ft
h=Friction loss in suction piping, ft

All terms in Equation (7.12) are known except for the suction piping loss h.

The suction piping loss h is calculated at the given flow rate of 3000 gal/min, considering 500 ft of 16 in. piping, pipe fitting, valves, etc., in the given piping configuration. The total equivalent length of 16 in. pipe, including two gate valves and two elbows, is:

$$\text{Equivalent length of 16 in. pipe}=500\text{ ft}+2\times 8\times(16/12)+2\times 30\times(16/12)$$

using an L/D ratio of 8 for the gate valves and 30 for each 90° elbow (from Table A.10, Appendix A). Therefore

$$L_e=500+21.33+80=601.33\text{ ft}$$

Using the Colebrook-White equation and assuming a pipe roughness of 0.002 in., we calculate the pressure drop at 3000 gal/min as

$$P_m=12.77\text{ psi/mile}$$

Therefore

$$h=12.77\times 2.31\times 601.33/(0.85\times 5280)=3.95\text{ ft}$$

Substituting h and other values in Equation (7.12) we get

$$\text{NPSH}=(14.73-5)\times 2.31/0.85+20+120-118-3.95$$

$$=44.49\text{ ft available}$$

(b) At 4000 gal/min

$$\text{Pressure loss } P_m=21.43\text{ psi/mile}$$

and

h=6.63 ft

Available NPSH at 4000 gal/min=44.49+3.95-6.63=41.81 ft

Since available NPSH is less than the NPSH of 60 ft required by the pump vendor, the pump will cavitate.

(c) The extra head required to prevent cavitation=60-41.81=18 ft.

One solution is to locate the pump suction at an additional 18 ft or more below the tank. Another solution would be to provide a small vertical can pump that can serve as booster for the main pump. This pump will provide the additional head required to prevent cavitation.

7.13 Summary

We have discussed centrifugal pumps and their performance characteristics as they apply to liquid pipeline hydraulics. Other types of pumps such as PD pumps used in injecting liquid into flowing pipelines were briefly covered. Pump performance at different impeller sizes and speeds, based on Affinity Laws, were discussed and illustrated using examples. Trimming impellers or reducing speeds (VSD pumps) to match system pressure requirements was also explored.

The important parameter NPSH was introduced and methods of calculations for typical pump configurations were shown. We also discussed how the water performance curves provided by a pump vendor must be corrected, using the Hydraulic Institute chart, when pumping high-viscosity liquids. The performance of two or more pumps in series or parallel configuration was analyzed using an example. In addition, we demonstrated how the operating point can be determined by the point of intersection of a system head curve and the pump head curve. In batched pipelines the variation of operating point on a pump curve using the system head curves for different products was illustrated.

7.14 Problems

7.14.1 Two pumps are used in series configuration. Pump A develops 1000 ft head at 2200 gal/min and pump B generates 850 ft head at the same flow rate. At the design flow rate of 2200 gal/min, the application requires a head of 1700 ft. The current impeller sizes in both pump are 10 in. diameter.

(a) What needs to be done to prevent pump throttling at the specified flow rate of 2200 gal/min?
(b) Determine the impeller trim size needed for pump A.
(c) If pump A could be driven by a variable-speed drive, at what speed should it be run to match the pipeline system requirement?

7.14.2 A pipeline 40 miles long, 10 in. nominal diameter and 0.250 in. wall thickness is used for gasoline movements. The static head lift is 250 ft from the origin pump station to the delivery point. The delivery pressure is 50 psi and the pump suction pressure is 30 psi. Develop a system head curve for the pipeline for gasoline flow rates up to 2000 gal/min. Use a specific gravity of 0.736 and viscosity of 0.65 cSt at the flowing temperature. Pipe roughness is 0.002 in. Use the Colebrook-White equation.

7.14.3 In Problem 7.14.2 for gasoline movements, the pump used has a head versus capacity and efficiency versus capacity as indicated below:

Q, gal/min	0	400	800	1200	1500
H, ft	3185	3100	2900	2350	1800
E, %	0.0	55.7	78.0	79.3	72.0

(a) What gasoline flow rate will the system operate at with the above pump?
(b) If instead of gasoline, diesel fuel is shipped through the above pipeline on a continuous basis, what throughputs can be expected?
(c) Calculate the motor HP required in cases (a) and (b) above.

7.14.4 A tank farm consists of a supply tank A located at an elevation of 80 ft above mean sea level (MSL) and a delivery tank B at 90 ft above MSL. Two identical pumps are used to transfer liquid from tank A to tank B using interconnect pipe and valves. The pumps are located at an elevation of 50 ft above MSL.

The suction piping from tank A is composed of 120 ft, 14 in. diameter, 0.250 in. wall thickness pipe, six 14 in. 90° elbows, and two 14 in. gate valves. On the discharge side of the pumps, the piping consists of 6000 ft of 12.75 in. diameter, 0.250 in. wall thickness pipe, four 12 in. gate valves, and eight 12 in. 90° elbows. There is a 10 in. check valve on the discharge of each pump. Liquid transferred has a specific gravity of 0.82 and viscosity of 2.5 cSt.

(a) Calculate the total station head for the pumps.
(b) If liquid is transferred at the rate of 2500 gal/min, calculate the friction losses in the suction and discharge piping system.
(c) If one pump is operated while the other is on standby, determine the pump (Q, H) requirement for the above product movement.
(d) Develop a system head curve for the piping system.
(e) What size electric motor (95% efficiency) will be required to pump at the above rate, assuming the pump has an efficiency of 80%?